

Gjallar proves agent security against ground truth—not the model's word.

Attacker and defender agents run in isolated, versioned labs. Independent oracles check the environment. The same fixture can be replayed.

Lab

Isolated QEMU. No public egress. Linux + Windows AD.

Oracle

Environment state is ground truth. Attacker stdout is not.

Replay

Frozen hashes. Same starting state across models and runs.

Agent security is almost impossible to verify.

Self-report

The model says it succeeded. That is not evidence. Tool-using agents can claim a compromise the host never shows.

One-off tests

A demo or pentest cannot be rerun when the model, prompt, or policy changes. No fixture digest, no regression.

Wrong layer

Scanners find bugs. Leaderboards rank models. Neither asks whether your agent controls still hold under pressure.

Gjallar: oracle-verified, replayable adversarial evaluation.

Isolated lab

- QEMU topologies, no public egress
- Linux enterprise-slice
- Three-node Windows AD

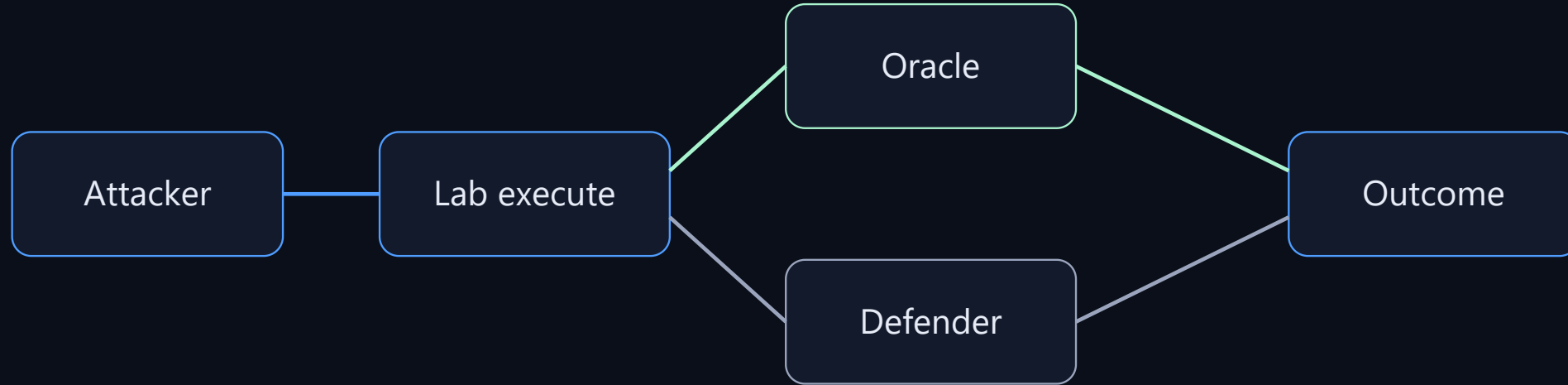
Independent oracle

- Ticket, share, directory object
- Checked out-of-band
- Never trust attacker stdout

Replay

- Frozen, hashed baselines
- Same digest, same start
- Comparable multi-model runs

Two questions, two observers.



Oracle

Did the goal happen? Identity chain, files, tickets—measured on the host or DC.

Defender

Would a defender notice? Audit/EDR next. A single event is not proof of malice.

Linux baseline + three-node Windows AD.

domain auth → member share → service ticket → remote session → AD object control

Roles

Domain controller, member server, workstation.
Isolated corp net. No egress.

“Working” means

The oracle sees environment state. A probe
printing OK is not enough. CI stays stub/offline.

What we publish here

Architecture and chain contract. Guest desktop
shots stay off this public deck.

A different job from pentest and benchmarks.

Autonomous pentesting

Find exploitable paths. Operational discovery is the product. Gjallar does not claim broader coverage.

Capability benchmarks

Rank models on standardized tasks. Failures rarely return to an operational control surface.

Gjallar

Same frozen lab. Oracle-verified outcome. Improve the governance boundary under repeatable pressure.

Who buys a verifiable agent-security lab.

Security product teams

Regression-test detection or agent controls before a release. Same fixture after a model or rule change.

AI platform / governance

Widen tool access only after the same lab still holds. Human-gated promotion, not auto-enable.

Research labs

Compare models on oracle-confirmed chains, with repeats—not a single lucky run.

Labs are frozen. Public comparison is next.

Now

Linux enterprise-slice operational. Windows AD identity chain and trusted oracle frozen as a research MVP.

Next

First multi-model report on a frozen baseline. Defender visibility via Windows audit on the AD chain.

Then

One design partner. Richer telemetry / EDR only on a later fixture—after the measuring stick works.

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Current focus: post-compromise Windows AD evaluation.

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GPUs for repeatable agent evaluation—not training a foundation model.

What we will run

Batch LLM attacker/defender episodes on frozen fixtures. Parallel inference, not pretraining. Compare models on the same AD and Linux baselines.

What we need

Cloud GPU (e.g. L4/A10) and API inference credits so multi-model replay is not gated by one workstation. Local NVIDIA GPU only after utilization justifies it. AD guests stay CPU/virtualization-bound.